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Innovation of Referencing Hallucination Score for medical AI Chatbots' and Comparison of Six Large Language Models

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Abstract

Background: Artificial intelligence (AI) chatbots have gained use recently in medical practice by healthcare practitioners. Interestingly, their output was found to have varying degrees of hallucination in content and references. Such hallucinations generate doubts about their output and their implementation.

Objective: We propose a reference hallucination score (RHS) to evaluate AI chatbots' citation authenticity.

Methods: Six AI chatbots were challenged with the same ten medical prompts, requesting ten references per prompt. The Reference Hallucination Score (RHS) is composed of six bibliographic items and the reference's relevance to prompts' keywords. RHS was calculated for each reference, prompt, and type of prompt (basic versus complex). The average RHS was calculated for each AI chatbot and compared across the different types of prompts and AI chatbots.

Results: Bard failed to generate any references. ChatGPT 3.5 and Bing generated the highest RHS (11), while Elicit and SciSpace generated the lowest RHS, and Perplexity was in the middle. The highest degree of hallucination was observed for reference relevancy to the prompt keywords (61.6%), while the lowest was reference titles (33.8%). AI chatbots generally had significantly higher RHS when prompted with scenarios or complex format prompts.

Conclusions: The variation in RHS underscores the necessity for a robust reference evaluation tool to improve the authenticity of AI chatbots. Also, it highlights the importance of verifying their output and citations. Elicit and SciSpace had negligible hallucination, while ChatGPT and Bing had critical levels. The proposed AI chatbots' RHS could contribute to ongoing efforts to enhance AI's general reliability in medical research.

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Innovation of Referencing Hallucination Score for medical AI Chatbots' and Comparison of Six Large Language Models

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Abstract

Background
Artificial intelligence (AI) chatbots have gained use recently in medical practice by healthcare practitioners. Interestingly, their output was found to have varying degrees of hallucination in content and references. Such hallucinations generate doubts about their output and their implementation. We propose a reference hallucination score (RHS) to evaluate AI chatbots' citation authenticity.

Methods
Six AI chatbots were challenged with the same ten medical prompts, requesting ten references per prompt. The Reference Hallucination Score (RHS) is composed of six bibliographic items and the reference's relevance to prompts' keywords. RHS was calculated for each reference, prompt, and type of prompt (basic versus complex). The average RHS was calculated for each AI chatbot and compared across the different types of prompts and AI chatbots.

Results
Bard failed to generate any references. ChatGPT 3.5 and Bing generated the highest RHS (11), while Elicit and SciSpace generated the lowest RHS (1), and Perplexity stood in the middle (7). The highest degree of hallucination was observed for reference relevancy to the prompt keywords (61.6%), while the lowest was reference titles (33.8%). ChatGPT and Bing had comparable RHS (Beta coefficient –0.069 *P* .320), while Perplexity had significantly lower RHS compared to ChatGPT (Beta coefficient –0.345 *P* <.001AI chatbots generally had significantly higher RHS when prompted with scenarios or complex format prompts (Beta coefficient 0.486, *P* <.001).

Conclusions
The variation in RHS underscores the necessity for a robust reference evaluation tool to improve the authenticity of AI chatbots. Also, it highlights the importance of verifying their output and citations. Elicit and SciSpace had negligible hallucination, while ChatGPT and Bing had critical levels. The proposed AI chatbots' RHS could contribute to ongoing efforts to enhance AI's general reliability in medical research.

Keywords
AI chatbots; Reference Hallucination; Bibliographic Verification; ChatGPT; Perplexity; SciSpace; Elicit; Bing

Introduction:

Artificial intelligence (AI) evolved from the early twentieth century until Alan Turing conceptualized machine learning in the 1950s and introduced the idea of using machines to process information in order to solve problems and make decisions [1]. Since then, scientists and others have pursued the dream of machines mimicking human intelligence through cognitive processes like thinking, data processing, and planning. AI development required foundational work in spoken language processing and information storage. Achieving these milestones faced numerous obstacles, including significant technical and financial challenges, which delayed AI's maturation.

A major hurdle in AI development was natural language processing (NLP), which progressed through multiple phases and culminated with the invention of bidirectional encoder representations from transformers (BERT), self-attention, and sequence-to-sequence deep learning technologies. These transformers marked a pivotal advancement in AI development, generating fluent and coherent large language models (LLMs), as they enabled the analysis of each word in an input sequence in context to its neighbors on both sides.

In 2018, OpenAI unveiled its first Generative Pre-trained Transformer, ChatGPT, enhancing its utility and potential applications in various human activities, including the healthcare and medical research sectors. Initially, ChatGPT stored data in its database without access to continually updated human literature. It later evolved to employ internet access. Subsequently, other AI chatbots specializing in healthcare and medical research were introduced. The implementation of AI chatbots in healthcare is optimistically viewed as a means to improve healthcare systems, medical education, and patient outcomes [2-4]. They are also viewed as potentially valuable tools in medical research and manuscript writing due to their capabilities in organization, data processing, text generation, and summarization [5, 6].

The medical literature surrounding AI has expanded tremendously since ChatGPT was publicly launched at the end of 2022 [7]. Most manuscripts have addressed its potential contributions to medical research and healthcare practices. However, a critical analysis of the literature reveals that a minority of AI chatbot users realized early that these LLMs do not voluntarily cite credible references to authenticate their information [8-10]. Even when ChatGPT and other AI chatbots were challenged to authenticate their outputs with references [11, 12], they generated multiple citations with detailed bibliographic data that seemed perfectly authentic, but most were actually fabricated, contained at least one falsified citation data, or were completely erroneous when verified through medical literature resources [11-14].

These fictitious citations raised major concerns about the AI chatbots' algorithms and methodology of NLP, especially regarding references, their bibliographic data credibility, and authenticity. Such erroneous outputs were described in the literature as AI hallucinations or fabrications. The mechanism of these fabrications remains unclear, though some AI developers have identified this faulty output and described it as a misalignment between user expectations and AI chatbot capabilities related to machine learning training issues without proposing definite explanation for that phenomenon or a methodology to assess its references' hallucination degree or impact on AI platforms output [15]. Another possible explanation for reference hallucinations is an encoding/decoding transformer glitch or the way AI chatbots handle bibliographic data as text amenable to manipulation. Some investigators assessed the artificial intelligence-generated medical information including ability to verify references and its publication date but did not pay attention specifically to certain identifiers of references that help to verify their authenticity [16].

Investigating the sources of these hallucinations and taking serious steps to fix contributing factors to this phenomenon is an urgent need in the current stage to implement AI technology in healthcare practice and medical research with high credibility. The first step stands by identifying the degree of these hallucinations especially in relation to references and their bibliographic data across the different AI chatbots.

Utilization of AI generated medical information is becoming a routine tool for physicians and trainees, medical education wise and even in fields of diagnosis and management. Therefore, the verification of this information references is extremely important to avoid adopting or utilizing erroneous unverified information and might be disastrous if used in medical management. Therefore, we propose to construct a reference evaluation tool that can universally be applied to assess AI chatbots generated references, which would be a helpful tool in assessing their output, its credibility, and its ability to be implemented into practice. This study aims to introduce this tool as "Reference Hallucination Score" (RHS), and to demonstrate its application in evaluating and comparing the outputs of six AI chatbots, stratify their citation output according to the (RHS), and assess the variables that are associated with Reference Hallucination Score. This initiative is the first to specifically address the gap in evaluating AI referencing hallucinations, offering a unique approach that fills a critical gap in enhancing the reliability of AI-generated references, thereby improving trust in AI outputs within medical research and guiding the development of more robust and reliable AI models.

Methods

Medical prompts:

We challenged six AI chatbots with ten medical prompts addressing various medical topics. The selected AI chatbots were ChatGPT 3.5, Bard, Perplexity, Bing, Elicit, and SciSpace [17-22]. The authors chose them because they are the most widely used AI chatbots in the literature and medical research and because of their ease of access by users. Using the focus group technique, the authors structured ten medical prompts with the best possible textual format and clarity to be understood by the AI chatbots. Each of the two prompts addressed a similar medical topic, the first in a basic format and the second in a complex or scenario-based format. The prompts are attached in the appendix. The five general topics for the medical prompts were glucose control in gestational diabetes, triggering factors in elderly asthmatic population, septic shock in infants with severe combined immunodeficiency (SCID), arthritis in patients with sickle cell disease, and substance abuse in patients with personality disorder.

All the prompts were formulated according to the following format: (1) Begin by searching PubMed for articles related to (topic); (2) Review the search results and select ten relevant and recent articles; (3) Ensure that all information is accurate and up-to-date; (4) Format the list of articles in a clear and organized manner, using a consistent style for each entry; (5) Include any additional information or notes that may be relevant or helpful for readers; and (6) Double-check the accuracy and completeness of the list before publishing or submitting it. The same ten prompts were applied to each of the six AI chatbots. Each prompt requested the AI chatbots to compile a list of ten references. Each should contain seven items for every reference: the reference title, journal name, authors, DOI, publication date, the article web link, and the reference's relevance to the keyword prompts.

Reference Hallucination Score:

Using the Delphi technique, we proposed the RHS (Table 1). Score development went through robust and meticulous methodology as following:

1. Literature Review: We commenced with an exhaustive review of existing literature to identify unique identifiers for reference scoring. This step was vital for understanding the current landscape and ensuring that our tool addresses the new gap in the field.
2. Expert Consultation Using the Delphi Technique: Our approach utilized the Delphi technique, a systematic, multi-step process involving rounds of anonymous feedback from experts to reach a consensus. We detail the implementation of this technique as follows:

3. Initial Survey of Senior Librarians: We consulted two senior librarians to gather insights on the proposed bibliographic identifiers, seeking their expert suggestions for improvements.
4. Expanded Survey of Senior Physicians: We further extended our consultation to 12 senior physicians and 11 junior physicians, most of whom are academics, to assess the relevance and importance of the suggested bibliographic identifiers, and to collect additional suggestions.
5. Consensus Building Among Authors: The final step involved synthesizing the feedback received and reaching a consensus among the authors, based on the mean results from the surveyed academics, to finalize the bibliographic identifiers for the hallucination score.

The RHS is an AI chatbot scoring system to evaluate article references generated by the AI chatbot based on the hallucination severity. The RHS is calculated based on the total score according to the presence of hallucination in any of the seven parameters. The parameters are four reference items given a score of 2 if they encountered any error in the reference title, journal name, authors' names, or DOI, as the authors judged it a major degree of hallucination. A score of 1 was given to any error in any of the other three parameters, which were reference publication date, reference web link, or reference relevance to the keyword prompts, as they were judged as minor degrees of hallucination. The maximum RHS score is 11, indicating the maximum degree of reference hallucination, and the minimum RHS score is 0, denoting no reference hallucination. To achieve the best outcome from the proposed RHS, the prompts submitted to the AI chatbot should include clear instructions to include all seven referencing items. The authenticity of the citations' items is evaluated by comparing AI chatbot responses to PubMed and Google Scholar responses. If the AI chatbot could not produce any reference to a specific prompt, the AI chatbot was given a score "N" for that prompt and was scored as failing to generate a result.

Table 1: Reference Hallucination Score (RHS)

Reference Items	**Item Hallucination Score**
1. Erroneous date of publication.	1
2. Erroneous web link of the article.	1
3. Erroneous citation relevance.	1
4. Erroneous title of the article.	2
5. Erroneous DOI.	2
6. Erroneous authors' names.	2
7. Erroneous name of the journal.	2
Total Reference Hallucination Score (RHS)	**11**

1. Erroneous date of publication: The publication date is missing or inaccurate.
2. Erroneous web link of the article: The link to the article is missing or directs to a different article or an error page.
3. Erroneous citation relevance: The keyword prompts are not in the title, abstract, or reference keywords.
4. Erroneous title of the article: The title provided by the AI platform is misspelled, incomplete, or nonexistent.
5. Erroneous DOI: DOI is missing, inaccurate, nonexistent, or directs to a different article.
6. Erroneous authors' names: Any author's name is missing, misspelled, or nonexistent.
7. Erroneous journal name: The journal's name is missing, misspelled, did not publish the

article, or does not exist.

Testing AI chatbots:

Our methodology of judging the hallucinations depended on using a systematic verification process based on PubMed and Google Scholar to ensure legitimacy, accuracy, and transparency. Each reference was initially verified using PubMed by searching for its DOI. If the DOI was inconclusive or unavailable, we leveraged a combination of the article's title, author names, and other vital details to ascertain its existence and accuracy within the PubMed database. In case of failure to reach the reference through PubMed, Google Scholar was the second step for verification using DOI, the article's title, authors, and other pertinent details to ensure the reference authenticity.

Each reference's total hallucination score was calculated according to the RHS methodology. Then, the mean RHS of the produced references of each prompt was calculated. This was applied to the six AI chatbots. The chatbots' RHS items were compared across the studied chatbots according to hallucination, correct results, and failure to generate results. The median RHS of complex prompts was compared with basic prompts, and finally, the median RHS of each AI chatbot was compared across all the studied chatbots. Linear regression was used to assess the independent association between mean RHS and other predictors, namely studied chatbot, prompt type, and prompt iteration.

Statistical analysis: The mean and standard deviation were used to describe continuous variables. The median and the interquartile range were used to describe continuous variables with statistical evidence of skewness. The frequency and percentage were used to describe categorically measured variables. The Kolmogrove-Smirnove statistical test and the histograms were used to assess the statistical normality assumption of metric variables. The Categorical Cronbach's alpha test was used to assess the internal consistency for reliability of the seven parameters of the RHS.

The Non-parametric Kruskall-Wallis ANOVA test was used to compare the RHS item results across the studied chatbots and to compare the median RHS for statistically significant differences. The Mann-Whitney U non-parametric test compared the chatbots' median RHS for statistically significant differences between basic and advanced prompts. The Multivariable Generalized Linear Models with Gamma Regression (GLMixed) analysis assessed significant differences in mean RHS by regressing it against the AI chatbot, prompt complexity, and prompt iterations. The association between mean RHS and tested predictor independent variables was expressed as an exponentiated beta coefficient (Risk Rate) with its associated 95% confidence interval. The SPSS IBM statistical computing software was used for the statistical data analysis, and the alpha significance level was considered at 0.050.

Results

A total of ten prompts were entered into each AI chatbot. Half inquired about basic medical topics, and half about clinical scenarios or complex medical topics. Each prompt requested ten references related to the prompt topic with their reference data according to our research methodology. Bard was the only AI chatbot that failed to produce any response to all ten applied medical prompts. Bard's response to our prompts was, "I'm a language model and don't have the capacity to understand and respond".

The AI chatbots failed to generate any response for 35 (7%) of the references. The highest hallucination/erroneous output was for the reference relevancy to the prompt content (61.6%), followed by reference web link (55.6%) and publication date (47.4%). Regarding the reference title and journal name, the AI chatbots' output had hallucination results of 33.8% and 37.6%, respectively.

Figure 1 displays each AI chatbot's correct and hallucination reference results. SciSpace and Elicit had the highest correct results of 629 and 597, respectively. ChatGPT had the highest hallucination results, at 592. Bing had the highest rate of failure to generate results, at 210.

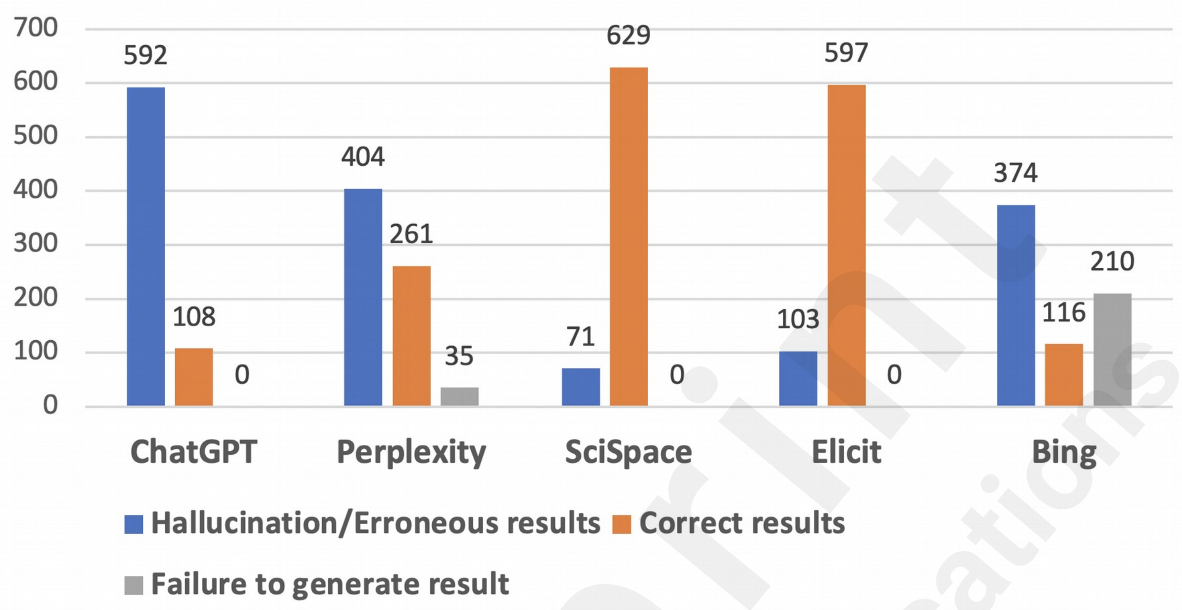

Figure 1: Individual AI chatbots' reference results, including hallucination, correct, and failure of generation.

Kruskall-Wallis non-parametric test showed that the AI chatbots differed significantly with respect to their total hallucination results ($\chi2(4)$ =205.9, *p <.001*), correct results ($\chi2(4)$ =305.02, *p <.001*) and failure to generate results ($\chi2(4)$ =104.3, *p <.001*). The Bonferroni adjusted post-hoc pairwise comparison test was used to compare the chatbots. SciSpace and Elicit did not differ significantly with respect to their hallucination or correct results (*p=1*). Both had significantly fewer hallucinations and the highest correct results compared to the other chatbots (*p <.001*).

ChatGPT had the highest rate of hallucination results compared to all the other chatbots. Bing and ChatGPT had no significant difference in their correct results (*p=1*). Bing and Perplexity had no significant difference in their hallucination results (*p =1*) and were significantly superior to ChatGPT. Perplexity was superior to both regarding its correct results (*p=.002*), (*p =.004*). Bing had the highest rate of failing to generate results compared to the rest (*p <.001*), followed by Perplexity.

Table 2 displays the detailed results of the studied AI chatbots' references' characteristics. The Kruskall-Wallis non-parametric test showed that the AI chatbots differed significantly in their results. SciSpace and Elicit generated the lowest number of hallucination results regarding title, journal name, authors' names, DOI and references' web link compared to all the other AI chatbots (*p <.001*). However, Elicit and Perplexity had no significant difference in hallucinating the publication date results, and both hallucinated more than SciSpace.

Table 2: AI chatbot hallucinating/erroneous reference results according to reference items

	Title	**DOI**	**Journal name**	**Authors' names**	**Publication date**	**Reference weblink**	**Reference relevance to the topic prompt**
ChatGPT	82						
Perplexity	41						

SciSpace	0						
Elicit	0						
Bing	46						
Test statistic, p-value	χ2(4) =231.4, p<0.001	χ2(4) =261.26, p<0.001	χ2(4) =249.4, p<0.001	χ2(4) =341.1, p<0.001	χ2(4) =199.9, p<0.001	χ2(4) =231.4, p<0.001	χ2(4) =10.77, p-value=0.029

Perplexity, Bing, and ChatGPT had no significant difference regarding their hallucination results for DOI, authors' names, reference web links, and reference titles. Perplexity and Bing hallucinated significantly less than ChatGPT regarding the journal name. Perplexity hallucinated slightly less than ChatGPT regarding reference title. Bing had significantly fewer hallucination results for references' irrelevancy to the prompt medical topic than ChatGPT (p =*.045*). The remaining AI chatbots, including Bing, did not show a significant difference (p >*.050*).

Table 3 shows AI chatbots' RHSs with the descriptive analysis of the median and interquartile range (IQR) for each studied AI chatbot. A Kruskall-Wallis analysis showed that chatbots differed significantly with respect to their overall measured hallucination scores (χ2(4) =277.7, p <*.001*). ChatGPT and Bing had the highest median RHSs but did not differ significantly (p =*1*). SciSpace and Elicit had the lowest median RHSs compared to all the other chatbots (p <*.001*) and did not differ significantly (p =*1*). Perplexity and Bing did not vary significantly with respect to their median total RHS (p =*.190*). On the other hand, Perplexity had a considerably lower median total RHS than ChatGPT (p =*.003*). For the prompt complexity, the median RHSs did not differ significantly for each of the studied AI chatbots (p >*.050*).

Table 3: Reference hallucination score (RHS) with bivariate analysis of the AI chatbots

	Hallucination score Median (IQR)[a]			**Test statistic**	***p-value***
	Total RHS	**Basic Prompts RHS**	**Advanced prompts RHS**		
ChatGPT	11 (1)	11 (1)	11 (0.25)	Z=1.25, DF=100	.209
Perplexity	7 (5)	7 (8.25)	8 (5)	Z=0.837, DF=95	.403
SciSpace	1 (1)	1 (1)	1 (1)	Z=0.942, DF=100	.346
Elicit	1 (2)	1 (2)	1 (2)	Z=0.207, DF=100	.836
Bing	11 (6)	11 (6)	11 (5)	Z=0.207, DF=70	.836

a: Interquartile range

Table 4 shows the multivariable generalized linear models with Gamma regression of the mean total hallucination score based on the chatbot, prompt complexity level, and prompt type. ChatGPT and Bing had the highest mean total hallucination scores and did not differ significantly (p =*.320*). SciSpace had the lowest mean total hallucination score compared to ChatGPT (Beta coefficient –1.748, p <*.001*). Elicit had the second-lowest total mean hallucination score compared to ChatGPT (Beta coefficient=–1.63, p <*.001*). Perplexity had the third-lowest score compared to ChatGPT (Beta coefficient=–0.345, p <*.001*).

Table 4: Multivariable generalized linear mixed regression analysis of the AI chatbots' total hallucination score

	Beta Coefficient	Beta coefficient 95% CI		*p-value*
		Lower	Upper	
Intercept	2.142	1.997	2.288	<.001
Chatbot Machine Bing	–0.069	–0.206	0.067	.320
Chatbot Machine Elicit	–1.630	–1.769	–1.492	<.001
Chatbot Machine SciSpace	–1.748	–1.880	–1.617	<.001
Chatbot Machine Perplexity	–0.345	–0.510	-0.181	<.001
Prompt complexity Level Advanced	0.486	0.326	0.645	<.001
Prompt Number	0.018	–0.006	0.041	.136
Interaction effect Prompt Number versus Prompt complexity	–0.077	–0.108	–0.046	<.001

Dependent Outcome variable: Total (overall) Digital Hallucination score+1.
Probability distribution: Gamma link function: Log

The level of prompt complexity also significantly affected the hallucination score when compared across all the AI chatbots. The advanced prompts' mean total hallucination score was significantly higher than the basic ones (Beta coefficient=0.486, *p <.001*). The prompt topic did not correlate significantly with the AI chatbots' mean hallucination score (*p =.136*). However, the interaction term between the prompt medical topic and prompt complexity was statistically significant (Beta coefficient=–0.077, *p <.001*), indicating that some topics, when presented to the chatbots in a complex/scenario, resulted in significantly lower mean total hallucinations compared to a basic presentation of the same topic (*p <.001*). The mean hallucination score based on prompt topic and prompt complexity did not significantly interact with any specific AI chatbots studied (Figure 2).

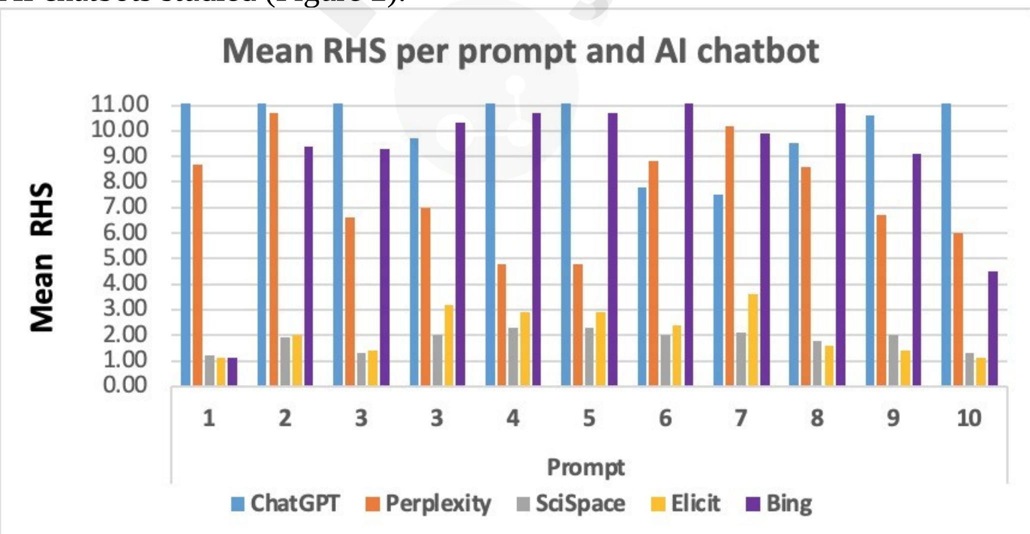

Figure 2: Mean RHS of each AI chatbot according to the prompt submitted.

Discussion

Our findings showed variation in the (RHS) across the different AI chatbots, ranging from almost null for SciSpace and Elicit to a critical, high degree of hallucination for ChatGPT[23]. Among the bibliographic items we studied, the publication date was the most common source of hallucination (59%), followed by the authors' names and DOI (about 45% and 47%, respectively). Reference relevancy to the topic prompt was the most common source of hallucination, reaching two-thirds on average, and 50% in Bing. Reference titles were the least source of hallucination 33.8%. Bard failed to generate any references for all the studied ten prompts.

The scientific community has used a transparent, reproducible, and accessible archiving system for its large, evolving research data. For example, FAIR guidelines ensure reference data is archived in a findable, accessible, interoperable, and reusable format to facilitate its citation, maintain researchers' credibility, and ensure data integrity and non-repudiation [24]. Citation is a vital step for information verification and authentication.

AI chatbots that have gained recent widespread use still lack a transparent and robust system for verifying citations and their information sources. Additionally, AI chatbots encounter a hallucination/fabrication phenomenon recognized early in their use in healthcare. Hallucinations have been encountered in various domains, including the content itself and the cited references, including their bibliographic identifiers [10, 25-27]. However, it is worth noting that this obstacle is improving gradually, especially by introducing research, dedicated medical chatbots and upgrading existing ones [14, 28].

A possible explanation for AI chatbots' referencing hallucination is the methodology LLMs use to handle citations. They may deal with them and their bibliographic identifiers as text, making them vulnerable to paraphrasing and other linguistic manipulations, and perhaps as a "watermark" so that the output that is generated by AI could be identified versus that is produced by humans, to reduce AI misuse [27, 29, 30]. Our colleague Buholayka et al pointed that ChatGPT is trained to give uninterrupted flow of conversation even at the cost of giving hallucinating results[31]. Another possible mechanism related to the AI chatbots' NLP methodology involves encoding and decoding defects during prompt processing, generating errors and fabricated results [32]. Additional factors might include insufficient training data for the LLM [15, 33, 34], context misinterpretation, lack of external validation, and an over-reliance on pattern recognition, all of which could contribute to variable degrees of hallucination, depending on the specific LLM and prompt structure.

Ye et al. constructed a systematic approach to AI chatbot hallucination by introducing a unique classification of hallucinations across diverse text generation actions, thereby furnishing theoretical insights, identification techniques, and enhancement strategies [35]. Their methodology consists of three domains: (1) comprehensive classification for hallucinations manifested in text generation tasks; (2) theoretical examinations of hallucinations in LLMs and amelioration; and (3) several research trajectories that hold promise for future exploration. Dhuliawala et al. suggested another model to potentially reduce hallucination by the Chain-of-Verification (CoVe) method [36]. Their four-step process consists of the chatbot drafting its initial response, formulating verification questions to scrutinize the draft, addressing these questions independently to prevent bias, and finally, generating a thoroughly verified response. Whether which model and which LLM gets a better hallucination score with time is yet to be determined in future research.

Our findings align with a study by Hua et al. that evaluated hallucinations in AI-generated ophthalmic scientific abstracts and references [23]. What makes our work unique, and pioneering is the introduction of a novel reference hallucination score to assess AI chatbots. It is based on six important reference items that make each reference unique and easily trackable for citation, in addition to a seventh item related to the relevance of the reference to the topic prompt [24]. The score was constructed based on any reference's most usable and unique items, with differential weights according to their uniqueness and importance in tracking and identifying any reference.

Variation in hallucination degree between the different bibliographic items highlights the variations among AI chatbots in handling and identifying references. Having a critically high hallucination rate of the cited reference’s relevance to the prompted topic stresses the possibility that AI chatbots identify certain keywords in the prompt and try to search for relevant references but, most of the time, fail to identify the correct and relevant keywords or as described by a recent Open AI company statement, they do not align properly with user intentions, leading to the citation of relatively irrelevant sources [15]. On the other hand, hallucinating article title, DOI are so risky in terms of inability to access the reference. Other identifiers hallucinations, like journal name or authors might not hugely block access to the cited reference.

According to our study, ChatGPT 3.5 had the highest total hallucination score, with the most hallucinations in all aspects of bibliographic items. This echoes others observation. Walters and Wilder described many hallucinations in ChatGPT’s bibliographic citations [14]. The study used ChatGPT 3.5 and ChatGPT 4 to generate literature reviews. It analyzed 636 bibliographic citations across 84 papers, finding a significant number of fabricated citations, 55% for GPT 3.5 and 18% for GPT 4, and errors in the non-fabricated ones, 43% for GPT 3.5 and 24% for GPT 4. Despite GPT 4 showing notable enhancement and insights over GPT 3.5, fabricated references persist [37]. Bing’s total hallucination score did not differ significantly from ChatGPT as they both had critically high scores. SciSpace and Elicit had comparable individual rates of hallucination regarding the different bibliographic items, although SciSpace had the lowest total hallucinations, followed by Elicit.

When considering the different studied reference items, ChatGPT, Bing, and Perplexity had comparable hallucinating results apart from the journal name, where Bing and Perplexity hallucinated less than ChatGPT. Perplexity stood in the middle, as its overall hallucination score was worse than SciSpace and Elicit on one side and better than ChatGPT and Bing on the other.

Our observations align with others who investigated AI platforms for writing and research objectives, as they found that Elicit and SciSpace are far more dedicated to searching for scientific papers and summarizing references [38]. This observation stresses scholars’ urge to vigilantly examine reference accuracy for any LLM-generated citations, especially if they are not dedicated to research purposes [10].

An interesting observation from our study is the failure of some chatbots to generate the prompted citations, as ChatGPT, Bing, and Perplexity had comparable individual hallucination reference results apart from journal names, where Bing and Perplexity hallucinated less than ChatGPT. Bing had a significantly higher rate of failure than ChatGPT, even though it performed comparably in all aspects. Such performance by Bing has been observed in another study that prompted ChatGPT, Bard, and Bing for MCQ generation, where Bing had a significant rate of generation failure compared to the other two [39].

Prompt structure and complexity had an interesting association with hallucination score, as complex or clinical scenarios triggered significantly more hallucinations across AI chatbots but not for certain ones. This was also observed when challenging ChatGPT versions 3.5 and 4 with orthopedic questions matched with images, as they both performed far better with simple text MCQs than those with images [40]. On the other hand, when prompted in a complex or scenario format, specific medical topics caused less hallucination than when prompted in a basic format. This observation might be explained by the transformer’s methodology and their differential performance with different text structures.

Bard performance might point to serious glitches in its text recognition or transformer performance in medical research, at least in certain topics or the ones used in our study. Previous work has shown serious fabrications encountered in Bard similar to ChatGPT [41].

Overall, our observations stream other findings that AI chatbots provided citations with varying degrees of inaccuracy or hallucination, necessitating users to independently verify the information obtained from these language models [41]. As such, hallucinations put the whole AI chatbot data output under significant question, especially in implementing and applying AI aid into clinical practice [42, 43].

Our study incorporated a meticulous design by constructing a novel scoring system to assess AI chatbots' hallucination in relation to references and their reference items with differential weights based on their importance. The multi-faceted verification approach we adopted is a robust method to ascertain the authenticity and relevance of references generated, providing a replicable model for future studies. That score has proven to differentiate the performance of five common chatbots skeptically. The invention of a hallucination score will be a vital step toward systematically evaluating and improving the referencing capabilities of AI chatbots and LLMs. It will also triage AI chatbots' hallucinations, which is a critical step in verifying the authenticity of their content.

Still, our study has potential limitations. The methodology used to construct the RHS is novel and is liable for future improvements. The RHS included a limited number of bibliographic parameters, which we believed based on our consensus and expert colleagues of academics, are the most important and unique references' identifiers. However, other researchers might perceive additional variables or identifiers as crucial. The prompt structure we used was after extensive trials to reach the prompt structure that produces AI generated references and their identifiers accurately as much as possible. Still, prompt design is liable for limitation, as LLM output depends hugely on the prompt structure fed to them. Prompt structure might explain partly the failure of Bard Chatbot to generate any references as it might need a special designs of prompting. However, we do not believe that the prompt strategy was a major limitation as it succeeded in generating output in almost all the other studied AI chatbots. Additionally, the medical prompts utilized might have a limited scientific scope, which may not cover the full spectrum of medical topics and scenarios that LLMs might encounter in real-world applications. Yet, proposed prompts to assess AI Chatbots referencing hallucination, require future refinement, especially with the introduction of new AI chatbots, specifically those specialized in the medical field.

Regarding references' identifiers verification process, even though we utilized multiple web-based steps utilizing PubMed and Google Scholar, this might still be suboptimal. Even though, albeit rigorous, relied heavily on existing databases and search engines, those engines might have their own set of limitations in indexing or recognizing all published literature, future researchers might propose more universal agreed on methodology in that regard. We selected only six chatbots to assess, this might not provide a fair and comprehensive understanding of the hallucination problems encountered across the myriad of AI chatbots that are increasingly becoming available and more specialized.

The potential biases in selecting AI chatbots and medical prompts, along with the verification process used, might impact the study results by not fully representing the breadth of AI capabilities and challenges. These limitations should be addressed in future research by expanding the number of chatbots studied, diversifying the medical prompts and their structure addressing medical basic knowledge, diagnosis and management, in order to strengthen the generalizability of the findings. Finally, refining the verification processes.

Our proposed RHS needs future sharpening and application to different and future AI chatbots, especially medical ones, to test its generalizability and sensitivity to assess references hallucination. Adding more bibliographic parameters might enhance its sensitivity. Furthermore, refining the definition of erroneous citations could potentially improve RHS performance and applicability, particularly in relation to the relevance of references to the prompted topic.

Conclusion

Our novel Reference Hallucination Score encompasses a methodology for delineating referencing inaccuracies, crucially in medical domains. It has shown variations across the analyzed AI chatbots. We evaluated the performance of six common AI chatbots. Elicit and SciSpace had the minimum hallucination, with almost none, while ChatGPT and Bing had a critical degree of hallucination. This emphasizes the pressing need for enhanced evaluation mechanisms of AI chatbots' output, particularly the cited references, and highlights the need to verify their output and apply it skeptically, all in order to grade AI chatbots credibility in terms of their contribution in healthcare and medical research areas. Additionally, our work establishes a foundation for ensuing research aimed at augmenting the reliability of AI chatbots in academic and clinical landscapes. Improving the LLM mechanism of references recognition and handling is an important necessity and needs maturation and improvement of the algorithms. Training in user prompt strategy is another trajectory to address to achieve the best performance of these chatbots and improve chatbots-user alignment. Future improvement of RHS or developing new versions will improve AI chatbots assessment and categorization and potentially will help artificial intelligence engineers to evaluate their work. The significance of the RHS and its potential impact on improving the reliability of AI-generated references cannot be overstated. The key takeaways highlight the broader implications for the use of AI in medical research, emphasizing the necessity for rigorous evaluations to enhance trust and reliability in AI outputs.

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Conceptualization, K.H.M., F.A. and M-H.T. Investigation, F.A., I.T., and K.H.M. Methodology, K.H.M., F.A. and M-H.T. Supervision, K.H.M., F.A. and M-H.T. Writing—original draft, F.A. and M-H.T. Writing—review & editing, F.A., M-H.T., I.T., A.A., A.J., K.A., T.A.M., M.F., and K.H.M. All authors have read and agreed to the published version of the manuscript.

Conflicts of Interest
The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Ethics approval:
This paper did not involve research on living creatures, so no IRB approval was required.

Data Availability:
The data may be made available upon reasonable request.

Appendix

Prompt 1 (Basic):

Act as an experienced researcher. Compile a list of ten recent articles on the variables that affect glucose control in gestational diabetes.

Include the following information for each article:

1. Article title.
2. Author(s).

3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 2 (Complex):

Act as an experienced researcher. Compile a list of ten recent articles on this question: Is an artificial pancreas or insulin pump effective in a second-trimester pregnant lady presenting with difficult-to-manage gestational diabetes?

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 3 (Basic):

Act as an experienced researcher. Compile a list of ten recent articles on the exacerbation triggering factors in elderly asthmatic population.

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.

5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 4 (Complex):

Act as an experienced researcher. Compile a list of ten recent articles on the best treatment regimen for poorly controlled asthma in the elderly with chronic rhinosinusitis and nasal polyps and congestive heart failure.

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 5 (Basic):

Act as an experienced researcher. Compile a list of ten recent articles on the causes of septic shock in infants with severe combined immunodeficiency (SCID).

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.

8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 6 (Complex):

Act as an experienced researcher. Compile a list of ten recent articles on this question: Is extracorporeal membrane oxygenation (ECMO) effective in pediatric patients with severe combined immunodeficiency (SCID), post bone marrow transplantation (BMT), admitted in PCIU with severe ARDS, graft-versus-host disease, septic shock, and multiorgan dysfunction syndrome?

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 7 (Basic):

Act as an experienced researcher. Compile a list of ten recent articles on the different mechanisms of arthritis in patients with sickle cell disease.

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.

2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 8 (Complex):

Act as an experienced researcher. Compile a list of ten recent articles on this case: A 17-year-old sickle cell disease patient has a history of recurrent episodes of Vaso occlusive crises and recurrent strokes, managed with monthly blood exchange transfusions for the last two years, presented with acute knee joint swelling. What are the possible pathological explanations?

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 9 (Basic):

Act as an experienced researcher. Compile a list of ten recent articles on the causes of substance abuse in patients with a personality disorder.

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each

entry.

5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

Prompt 10 (Complex):

Act as an experienced researcher. Compile a list of ten recent articles on the possible mechanisms of depression and anxiety symptoms in a patient admitted over ingestion of herbs and dietary supplements.

Include the following information for each article:

1. Article title.
2. Author(s).
3. Journal name.
4. Date of publication.
5. Number of citations.
6. DOI.
7. Web link to the article.
8. PubMed link.

Instructions:

1. Begin by searching PubMed.
2. Review the search results and select ten recent articles that are relevant.
3. Ensure that all information is accurate and up to date.
4. Format the list of articles in a clear and organized manner, using a consistent style for each entry.
5. Include any additional information or notes that may be relevant or helpful for readers.
6. Double-check the accuracy and completeness of the list before publishing or submitting it.

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Supplementary Files

Untitled.
URL: http://asset.jmir.pub/assets/51e16be1f8f6cf036cc83e570bbbd7c3.docx

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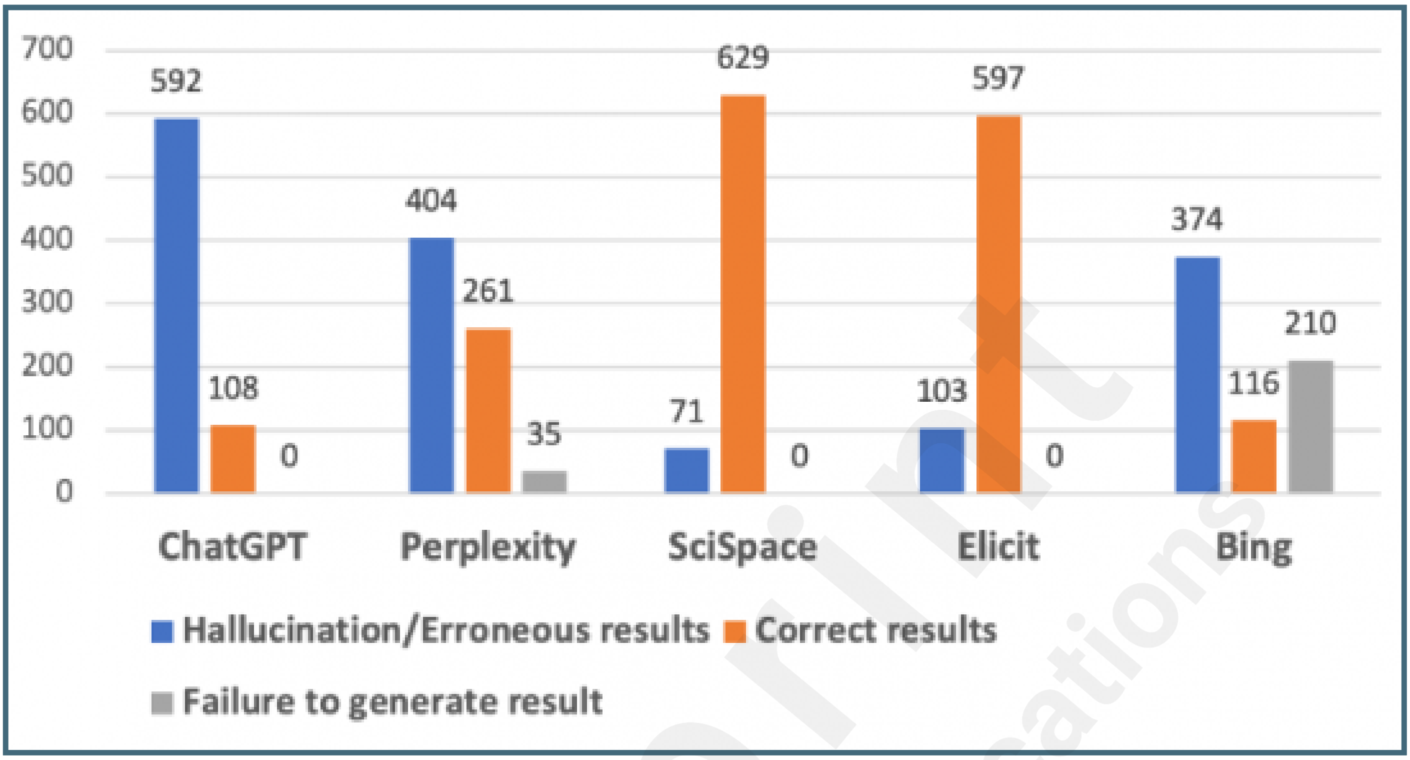
Untitled.
URL: http://asset.jmir.pub/assets/3077fd6d6b4111ffe2688f753f1bee4e.docx

Figures

Individual AI chatbots' reference results, including hallucination, correct, and failure of generation.



Mean RHS of each AI chatbot according to the prompt submitted.